MIL-W-13074A(EL)
19 MAY 1965

SUPERSEDING
MIL-W-13074(SigC)
14 October 1953

MILITARY SPECIFICATION

WIRE, ELECTRICAL, ANTENNA,
(WIRE W-27 AND WIRE WS-19( )/U)

1. SCOPE

1.1 This specification covers stranded, round electrical wire fabricated from silicon bronze, suitable for antenna applications.

1.2 Classification. - Silicon bronze wire covered by this specification shall be of the following types (see 6.2.):

1.2.1 Wire W-27 consists of 7 strands of 0.064 inch diameter silicon bronze wire approximately equivalent to 26,300 circular mils (No. 6 AWG) in area.

1.2.2. Wire WS-19( )/U consists of 7 strands of 0.040 inch diameter silicon bronze wire approximately equivalent to 10,400 circular mils (No. 10 AWG) in area. (See 6.3)

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-12000 Cable, Cord, and Wire, Electric; Packaging of.

STANDARDS

MILITARY

MIL-STD-105 Sampling Inspection Tables for Inspection by Attributes.

FSC 6145

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. - The material for the wire shall be as specified herein. The best material commercially available for the purpose shall be used when a definite material is not designated.

3.2 Strands. -

3.2.1 Each strand of wire W-27 shall consist of a silicon bronze alloy wire with a diameter of 0.064 inch, ±0.0015 inch.

3.2.2 Each strand of Wire WS-19( )/U shall consist of a silicon bronze alloy wire with a diameter of 0.040 inch, ±0.0015 inch.

3.3 Conductor construction. -

3.3.1 W-27. - Wire W-27 shall be constructed of 7 strands of wire as specified in 3.2.1, twisted with a left-hand lay of a maximum 3 inches in length.

3.3.2 WS-19( )/U. - Wire WS-19( )/U shall be constructed of 7 strands of wire as specified in 3.2.2 twisted with a left-hand lay of a maximum 2 inch in length.

3.4 Joints in conductors. - Joints in conductors should be avoided, and in no case shall they exceed two joints per strand per length of wire, with joints spaced not less than 3 feet apart. All joints shall be butt brazed using silver alloy solder and a nonacid flux; and shall be finished smoothly in a workmanlike manner. No twist joints shall be used. The tensile strength of a section of each strand which includes a joint shall not be less than 80 percent of the tensile strength of an adjacent section of the strand of equal length without a joint. The average tensile strength of a group of twelve (12) strands which include joints shall not be less than 90 percent of the adjacent sections of the strands.

3.5 Breaking strength. -

3.5.1 W-27. - Wire W-27 shall have a breaking strength of not less than 2200 pounds.

3.5.2 WS-19( )/U. - Wire WS-19( )/U shall have a breaking strength of not less than 790 pounds.

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3.6 Elongation. - The elongation shall be not less than 0.80 percent for a 60 inch length of stranded conductor.

3.7 Workmanship. - Strands and wire shall be manufactured and finished in a careful and workmanlike manner in accordance with the good design and sound practice.

Section 4 - QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling

4.2.1 For quality conformance tests, sampling shall be in accordance with Standard MIL-STD-105.

4.3 Classification of inspection. - The inspection and testing of wire W-27 and wire W8-19 shall be classified as follows:

(a) Operator qualification test.

(b) Quality conformance inspection before preparation for delivery.

(c) Quality conformance inspection of preparation for delivery.

4.4 Operator qualification test. - This test shall be performed at the start of production for each order for wire under this specification and shall be as follows:

Each operator required to make joints in strands will submit a group of twelve (12) specimens of strand which include a joint with a specimen of adjacent normal section of each. These specimens will be tested for breaking strength. An operator will be considered qualified after submitting a group of twelve specimens which meet the requirements of 3.4. Records of such qualifications shall be available for verification by the Government.

4.5 Quality conformance inspection before preparation for delivery. The supplier shall perform the inspection specified in 4.4, 4.5.1 and 4.5.2. This does not relieve the supplier of his responsibility for performing any additional inspection which is necessary to control the quality of the product

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and to assure compliance with all specification requirements. The Government will review and evaluate the supplier's inspection procedures and examine the supplier's inspection records. In addition the Government-at its discretion - may perform all or any part of the specified inspection, to verify the supplier's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection. - This inspection, including sampling, shall consist of visual and dimensional examinations of paragraph 4.6 to determine compliance with the requirements of paragraph 3.1 through 3.4 and 3.7 as applicable. The AQL shall be 1.0% for major defects and 4.0% for minor defects.

4.5.2 Group B inspection. - This inspection, including sampling, shall consist of the tests and examinations of paragraph 4.7. Sampling shall conform to the special level S-4 of MIL-STD-105 for normal, tightened and reduced inspection. AQL shall be 4.0%.

4.6 Visual and dimensional inspection. - Strands and completed wire shall be inspected to verify that the materials and all workmanship comply with this specification.

4.6.1 Construction and lay. - Construction and lay shall be determined by physical examination and measurement of the complete wire.

4.7 Breaking strength and elongation. - Breaking strength and elongation shall be made on a tensile strength machine, preferably of the pendulum types. Breaking strength test shall be made after stranding, on the completed wire. The free length between grips of the test specimen shall be not less than 60 inches and the speed shall be not more than 3 inches per minute. Care shall be taken to insure that strands in the stranded wire are evenly gripped during test. (Successful testing of stranded wire for tensile properties requires an adequate means of gripping the ends of the test specimen. Various means are available, such as a long tube or socket into which the wire may be soldered, or in which, after insertion, the wires may be swaged or pressed without serious distortion. Ordinary jaws or clamping devices are usually not suitable.) At the conclusion of the test, the wire shall meet the requirements for breaking strength and elongation specified in 3.5 and 3.6, respectively.

4.8 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Wires shall be preserved, packaged, packed and marked in accordance with Specification MIL-C-12000. (See 6.2).

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5.2 Preservation, packaging and packing shall be level A, B, or C as specified. (See 6.2).

6. NOTES

6.1 Intended use. - Wire covered by this specification is most commonly used in antenna application.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of wire and length required. (See 1.2)
(c) Size of reel required and length on each when applicable.
(d) Level of packaging and level of packing required. (See 5.1, 5.2).

6.3 Nomenclature. - The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: WS-19W/U. As soon as possible after the award of contract, the contractor should apply to the Government office specified in the contract for such information.

6.4 Definitions

6.4.1 Inspection lot. - An inspection lot shall consist of all units of product of one size, produced under essentially the same conditions and submitted for acceptance at the same time.

6.4.2 Unit of product. - A unit of product is the continuous length of wire contained on a single reel, or spool, or in a package.

6.4.3 Specimen. - A specimen consists of the individual piece of the unit of product upon which the test is made.

CUSTODIAN:

ARMY- EL

PREPARING ACTIVITY:

ARMY-EL

Project Nr. 6145-A037

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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